

COOCHBEHAR GOVERNMENT ENGINEERING COLLEGE

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EXPERIMENT 7 — PULSE SHAPING AND THE NYQUIST CRITERION

Digital Communication Laboratory

1. Aim

To study and compare rectangular, sinc, raised-cosine (RC), and root-raised-cosine (RRC) pulses; verify the Nyquist zero-crossing criterion; and study the effect of roll-off factor on bandwidth and time-domain pulse shape.

2. Objectives

- Generate an upsampled symbol impulse train.
- Generate and compare rectangular, sinc, raised-cosine and root-raised-cosine pulses.
- Test raised-cosine roll-off factors $\alpha = 0, 0.25, 0.5, 1$.
- Plot impulse/time-domain and frequency responses.
- Verify zero crossings at integer symbol intervals.
- Generate a pulse-shaped data stream.
- Cascade transmitter and receiver RRC filters.
- Study bandwidth versus roll-off factor.

3. Theory

3.1 Pulse Shaping

In digital communication, ideal impulses cannot normally be transmitted because they require infinite bandwidth. Pulse shaping converts each symbol into a suitable waveform before transmission. Its main purposes are to limit transmission bandwidth, reduce inter-symbol interference (ISI), and facilitate reliable symbol detection.

3.2 Nyquist Criterion

The Nyquist criterion provides the condition for zero inter-symbol interference. For a pulse $p(t)$, the condition is:

$$p(nT) = 1, n = 0$$

$$p(nT) = 0, n = \pm 1, \pm 2, \pm 3, \dots$$

where T is the symbol period. Thus, at every ideal sampling instant, only the desired symbol contributes and all neighboring symbols contribute zero.

4. Different Pulse Shapes

4.1 Rectangular Pulse

$p(t) = 1$ for $|t| < T/2$, and $p(t) = 0$ otherwise. It is simple to generate, but its frequency spectrum has large sidelobes and therefore significant out-of-band energy.

4.2 Sinc Pulse

$p(t) = \text{sinc}(t/T)$, where $\text{sinc}(x) = \sin(\pi x)/(\pi x)$. The sinc pulse has zeros at all integer multiples of T other than $t = 0$, so it satisfies the ideal Nyquist zero-ISI condition. However, it is infinitely long in time and is not directly realizable without truncation.

4.3 Raised-Cosine Pulse

The raised-cosine impulse response is:

$$h_{\text{RC}}(t) = \text{sinc}(t/T) \cdot \cos(\pi \alpha t/T) / [1 - (2\alpha t/T)^2]$$

where $0 \leq \alpha \leq 1$ and α is the roll-off factor. For $\alpha = 0$, the raised-cosine pulse reduces to the sinc pulse.

4.4 Roll-Off Factor

The roll-off factor determines the excess bandwidth. For symbol period T , the one-sided baseband bandwidth is:

$$B = (1 + \alpha)/(2T)$$

Roll-off α	Bandwidth
0	$1/(2T)$
0.25	$0.625/T$
0.5	$0.75/T$
1	$1/T$

Increasing α increases bandwidth, makes the pulse more localized in time, and generally improves tolerance to timing imperfections. This is the bandwidth–timing trade-off.

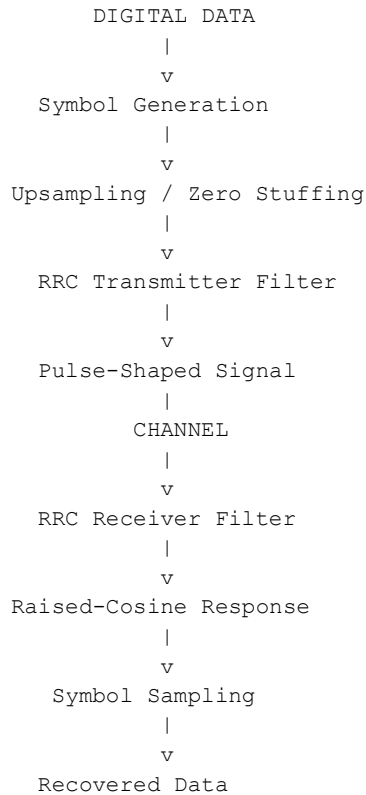
4.5 Root Raised-Cosine Filter

The root-raised-cosine filter is designed so that two identical RRC filters, one at the transmitter and one at the receiver, produce the overall raised-cosine response:

$$H_{\text{RRC}}(f) \times H_{\text{RRC}}(f) = H_{\text{RC}}(f)$$

Thus, $\text{RRC}_{\text{Tx}} \times \text{RRC}_{\text{Rx}} = \text{RC}$.

5. Block Diagram



6. Software Required

MATLAB with basic numerical and plotting functions.

7. MATLAB Program

```
clc;
clear;
close all;

%% EXPERIMENT 7
% Pulse Shaping and the Nyquist Criterion

%% Parameters
T = 1; % Symbol period
Rs = 1/T; % Symbol rate
sps = 100; % Samples per symbol
Fs = sps/T; % Sampling frequency
span = 10; % Pulse duration in symbols

t = -span*T : 1/Fs : span*T;

alpha_values = [0 0.25 0.5 1];

%% 1. Generate Rectangular, Sinc, Raised-Cosine and RRC Pulses

rect_pulse = double(abs(t) <= T/2);
sinc_pulse = sinc(t/T);

RC = zeros(length(alpha_values),length(t));
```

```

RRC = zeros(length(alpha_values),length(t));

for k = 1:length(alpha_values)

    alpha = alpha_values(k);

    %% Raised Cosine
    if alpha == 0
        RC(k,:) = sinc(t/T);
    else
        x = t/T;

        h = sinc(x).*cos(pi*alpha*x) ./ ...
            (1-(2*alpha*x).^2);

        % t = 0
        idx0 = abs(x) < 1e-12;
        h(idx0) = 1;

        % Removable singularities:
        % t = +/-T/(2*alpha)
        idx1 = abs(abs(x)-1/(2*alpha)) < 1e-12;
        h(idx1) = (alpha/2)*sin(pi/(2*alpha));

        RC(k,:) = h;
    end

    %% Root Raised Cosine
    if alpha == 0
        RRC(k,:) = sinc(t/T);
    else
        x = t/T;
        h = zeros(size(x));

        idx = (abs(x) > 1e-12) & ...
            (abs(abs(x)-1/(4*alpha)) > 1e-12);

        h(idx) = ...
            (sin(pi*x(idx)*(1-alpha)) + ...
            4*alpha*x(idx).*cos(pi*x(idx)*(1+alpha))) ./ ...
            (pi*x(idx).*(1-(4*alpha*x(idx)).^2));

        % t = 0
        idx0 = abs(x) < 1e-12;
        h(idx0) = 1-alpha+4*alpha/pi;

        % t = +/-T/(4*alpha)
        idx1 = abs(abs(x)-1/(4*alpha)) < 1e-12;

        h(idx1) = ...
            (alpha/sqrt(2))* ...
            ((1+2/pi)*sin(pi/(4*alpha)) + ...
            (1-2/pi)*cos(pi/(4*alpha)));

        RRC(k,:) = h;
    end
end

%% 2. Plot Pulse Responses
figure;
subplot(2,2,1);
plot(t,rect_pulse,'LineWidth',1.5);

```

```

grid on;
xlabel('Time (T)');
ylabel('Amplitude');
title('Rectangular Pulse');

subplot(2,2,2);
plot(t,sinc_pulse,'LineWidth',1.5);
grid on;
xlabel('Time (T)');
ylabel('Amplitude');
title('Sinc Pulse');

subplot(2,2,3);
plot(t,RC(2,:), 'LineWidth',1.5);
grid on;
xlabel('Time (T)');
ylabel('Amplitude');
title('Raised Cosine, \alpha = 0.25');

subplot(2,2,4);
plot(t,RRC(2,:), 'LineWidth',1.5);
grid on;
xlabel('Time (T)');
ylabel('Amplitude');
title('Root Raised Cosine, \alpha = 0.25');
sgtitle('Comparison of Pulse Shapes');

%% 3. Raised-Cosine Pulse for Different Roll-Off Factors
figure;
hold on;
for k = 1:length(alpha_values)
    plot(t,RC(k,:), 'LineWidth',1.5);
end
grid on;
xlabel('Time (T)');
ylabel('Amplitude');
title('Raised-Cosine Pulse for Different Roll-Off Factors');
legend('\alpha=0','\alpha=0.25','\alpha=0.5','\alpha=1');
xlim([-5 5]);

%% 4. RRC Pulse for Different Roll-Off Factors
figure;
hold on;
for k = 1:length(alpha_values)
    plot(t,RRC(k,:), 'LineWidth',1.5);
end
grid on;
xlabel('Time (T)');
ylabel('Amplitude');
title('Root Raised-Cosine Pulse for Different Roll-Off Factors');
legend('\alpha=0','\alpha=0.25','\alpha=0.5','\alpha=1');
xlim([-5 5]);

%% 5. Frequency Response
NFFT = 16384;

figure;
hold on;
for k = 1:length(alpha_values)
    H = fftshift(fft(RC(k,:),NFFT));
    f = (-NFFT/2:NFFT/2-1)*Fs/NFFT;
    Hmag = abs(H);

```

```

    Hmag = Hmag/max(Hmag);
    plot(f,20*log10(Hmag+1e-12),'LineWidth',1.2);
end
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Raised-Cosine Frequency Responses');
legend('\alpha=0','\alpha=0.25','\alpha=0.5','\alpha=1');
xlim([-1.5 1.5]);
ylim([-80 5]);

%% 6. Verify Nyquist Zero Crossings
alpha_test = 0.5;
x = t/T;

h_test = sinc(x).*cos(pi*alpha_test*x) ./ ...
    (1-(2*alpha_test*x).^2);

h_test(abs(x)<1e-12) = 1;

idx_sing = abs(abs(x)-1/(2*alpha_test)) < 1e-12;
h_test(idx_sing) = ...
    (alpha_test/2)*sin(pi/(2*alpha_test));

symbol_indices = -5:5;
zero_crossing_values = zeros(size(symbol_indices));

for n = 1:length(symbol_indices)
    target = symbol_indices(n)*T;
    [~,index] = min(abs(t-target));
    zero_crossing_values(n) = h_test(index);
end

disp(' ');
disp('Nyquist Zero-Crossing Validation');
disp('-----');
disp(' n          t=nT          RC response');

for n = 1:length(symbol_indices)
    fprintf('%3d      %7.3f      %+.6e\n', ...
        symbol_indices(n), ...
        symbol_indices(n)*T, ...
        zero_crossing_values(n));
end

figure;
stem(symbol_indices,zero_crossing_values,'filled');
grid on;
xlabel('Symbol Index n');
ylabel('h_{RC}(nT)');
title('Nyquist Zero-Crossing Verification');

%% 7. Generate Upsampled Symbol Impulse Train
data = [1 -1 1 1 -1 -1 1 -1];

upsampled = zeros(1,length(data)*sps);
upsampled(1:sps:end) = data;

time_data = (0:length(upsampled)-1)/Fs;

figure;
stem(time_data,upsampled,'filled');
```

```

grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Upsampled Symbol Impulse Train');

%% 8. Generate Pulse-Shaped Data Stream
alpha_stream = 0.5;
rrc_stream = RRC(3,:);
rrc_stream = rrc_stream / max(abs(rrc_stream));

tx_signal = conv(upsampled,rrc_stream,'same');

figure;
plot(time_data,tx_signal,'LineWidth',1.3);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Pulse-Shaped Data Stream using RRC');

%% 9. Cascade Transmitter and Receiver RRC Filters
cascade_rrc = conv(rrc_stream,rrc_stream);
cascade_rrc = cascade_rrc/max(abs(cascade_rrc));

rc_compare = RC(3,:);
rc_compare = rc_compare/max(abs(rc_compare));

t_cascade = ((0:length(cascade_rrc)-1) ...
    - floor(length(cascade_rrc)/2))/Fs;

figure;
plot(t_cascade,cascade_rrc,'LineWidth',1.5);
hold on;
plot(t,rc_compare,'--','LineWidth',1.5);
grid on;
xlabel('Time (T)');
ylabel('Amplitude');
title('Cascade of Tx-RRC and Rx-RRC');
legend('RRC * RRC','Raised Cosine');
xlim([-5 5]);

%% 10. Frequency Response of RRC
figure;
hold on;
for k = 1:length(alpha_values)
    H = fftshift(fft(RRC(k,:),NFFT));
    f = (-NFFT/2:NFFT/2-1)*Fs/NFFT;
    Hmag = abs(H);
    Hmag = Hmag/max(Hmag);
    plot(f,20*log10(Hmag+1e-12),'LineWidth',1.2);
end
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Root Raised-Cosine Frequency Responses');
legend('\alpha=0','\alpha=0.25','\alpha=0.5','\alpha=1');
xlim([-1.5 1.5]);
ylim([-80 5]);

%% 11. Bandwidth versus Roll-Off Factor
B = (1+alpha_values)/(2*T);

figure;

```

```

plot(alpha_values,B,'o-','LineWidth',1.5);
grid on;
xlabel('Roll-Off Factor \alpha');
ylabel('Bandwidth (Hz)');
title('Bandwidth versus Roll-Off Factor');

disp(' ');
disp('Bandwidth versus Roll-Off Factor');
disp('-----');
disp(' Alpha          Bandwidth');

for k = 1:length(alpha_values)
    fprintf(' %.2f          %.4f Hz\n', ...
        alpha_values(k),B(k));
end

disp(' ');
disp('Experiment completed successfully.');
```

disp('The RC pulse satisfies the Nyquist zero-ISI criterion.');

disp('Increasing roll-off factor increases bandwidth.');

disp('Two cascaded RRC filters produce an RC response.');

8. Step-by-Step Procedure

Step 1: Set $T = 1$ s, samples per symbol = 100, and generate the time vector.

Step 2: Generate rectangular and sinc pulse responses.

Step 3: Generate RC and RRC responses directly from their mathematical definitions.

Step 4: Handle the removable singularities at $t = 0$ and the special RC/RRC points explicitly.

Step 5: Test $\alpha = 0, 0.25, 0.5$ and 1 .

Step 6: Plot the time-domain pulse responses.

Step 7: Use FFT to calculate and plot frequency responses.

Step 8: Sample the RC response at integer symbol intervals and tabulate the values to verify the Nyquist zero crossings.

Step 9: Generate an upsampled symbol impulse train from the data sequence $[1 \ -1 \ 1 \ 1 \ -1 \ -1 \ 1 \ -1]$.

Step 10: Pass the upsampled sequence through an RRC filter to obtain the pulse-shaped stream.

Step 11: Cascade transmitter and receiver RRC filters and compare the result with the direct RC response.

Step 12: Calculate theoretical bandwidth $B = (1 + \alpha)/(2T)$ and plot bandwidth versus α .

9. Nyquist Zero-Crossing Validation

For $\alpha = 0.5$, the RC pulse should satisfy $h_{RC}(0) = 1$ and $h_{RC}(nT) \approx 0$ for $n \neq 0$.

n	t = nT	Expected $h_{RC}(nT)$
-5	-5T	≈ 0

-4	-4T	≈ 0
-3	-3T	≈ 0
-2	-2T	≈ 0
-1	-1T	≈ 0
0	0T	1
1	1T	≈ 0
2	2T	≈ 0
3	3T	≈ 0
4	4T	≈ 0
5	5T	≈ 0

Small numerical values such as 10^{-16} or 10^{-17} may appear instead of exact zero because of floating-point precision, finite pulse truncation, and discrete-time sampling.

10. Expected Observations

Roll-off α	Time-domain behavior	Bandwidth
$\alpha = 0$	Sinc-like, slow time-domain decay	Minimum bandwidth
$\alpha = 0.25$	Faster decay	Higher bandwidth
$\alpha = 0.5$	More localized pulse	Higher bandwidth
$\alpha = 1$	Fastest decay among tested cases	Maximum bandwidth

11. Interpretation

As α increases, the raised-cosine pulse becomes more localized in time while its frequency transition band becomes wider. Thus, increasing α increases bandwidth but reduces long time-domain tails.

The RC pulse has zero crossings at integer symbol intervals except at the main sampling point, confirming the zero-ISI Nyquist condition.

When an RRC filter is used at both transmitter and receiver, the cascaded response approaches the raised-cosine response, confirming the theoretical relationship $\text{RRC} \times \text{RRC} = \text{RC}$.

12. Bandwidth Calculation

For $T = 1$ s:

- $\alpha = 0$: $B = 0.50$ Hz
- $\alpha = 0.25$: $B = 0.625$ Hz
- $\alpha = 0.5$: $B = 0.75$ Hz
- $\alpha = 1$: $B = 1.00$ Hz

13. Error / Discrepancy Analysis

If the simulated zero crossings are not exactly zero, the discrepancy is normally due to floating-point numerical precision, finite-duration pulse truncation, and discrete-time sampling. The diagnostic test used is direct sampling of the RC response at $t = nT$ and comparison against the expected Nyquist values.

14. Result

The experiment successfully demonstrates generation and comparison of rectangular, sinc, raised-cosine and root-raised-cosine pulses; the effect of roll-off factor; frequency-domain behavior; Nyquist zero-crossing verification; pulse shaping of a digital data stream; RRC transmitter/receiver cascading; and bandwidth versus roll-off factor.

15. Conclusion

The pulse-shaping experiment was successfully performed using MATLAB. The raised-cosine pulse satisfies the Nyquist zero-ISI criterion because its response is zero at all integer symbol intervals except the desired sampling instant. Increasing the roll-off factor increases the required bandwidth but gives better time-domain localization. Two cascaded root-raised-cosine filters produce an overall raised-cosine response, making RRC filtering highly suitable for practical digital communication systems.

Key relations:

- Higher $\alpha \rightarrow$ higher bandwidth but better time localization.
- $\text{RRC_Tx} \times \text{RRC_Rx} = \text{RC}$.
- $h_{\text{RC}}(nT) = 0$ for $n \neq 0$.
- $B = (1 + \alpha)/(2T)$.

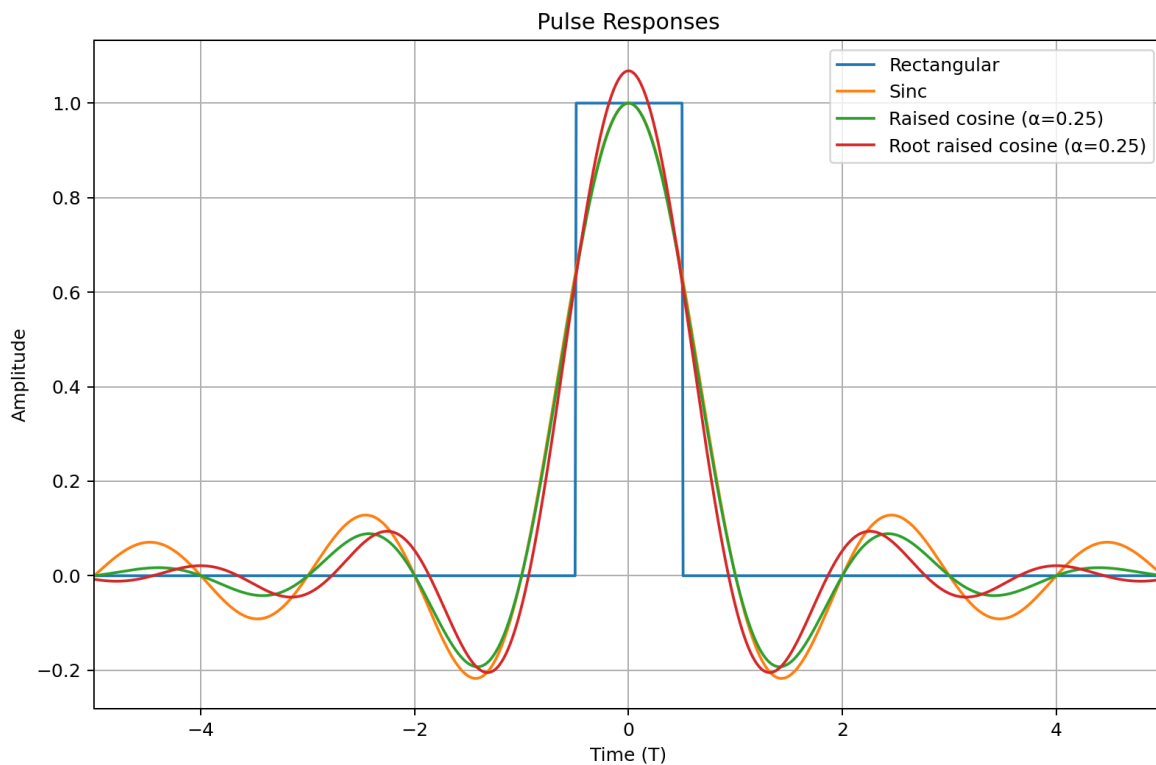
16. Viva Questions and Answers

Question	Answer
What is pulse shaping?	Pulse shaping converts digital symbols into suitable pulses to control bandwidth and reduce ISI.
What is ISI?	Inter-Symbol Interference is interference in which one symbol affects neighboring symbols.
What is the Nyquist criterion?	It is the condition that the pulse has zero amplitude at all other symbol sampling instants.
What is roll-off factor?	It is the parameter α that determines the excess bandwidth of a raised-cosine filter.
What is the range of α ?	$0 \leq \alpha \leq 1$.
What happens when $\alpha = 0$?	The raised-cosine pulse becomes an ideal sinc pulse.
What happens when α increases?	Bandwidth increases and the pulse becomes more localized in time.
Why is RRC used?	It divides the overall raised-cosine filtering between transmitter and receiver.
What happens when two RRC filters are cascaded?	The overall response becomes a raised-cosine response.
What is the RC bandwidth?	$B = (1 + \alpha)/(2T)$.
Why are zero crossings important?	They ensure that neighboring symbols contribute zero at the sampling instant, reducing ISI.
What is the bandwidth–timing trade-off?	Lower α saves bandwidth, while higher α improves time localization and timing tolerance at the cost of bandwidth.

17. Required Graphs / Simulation Results

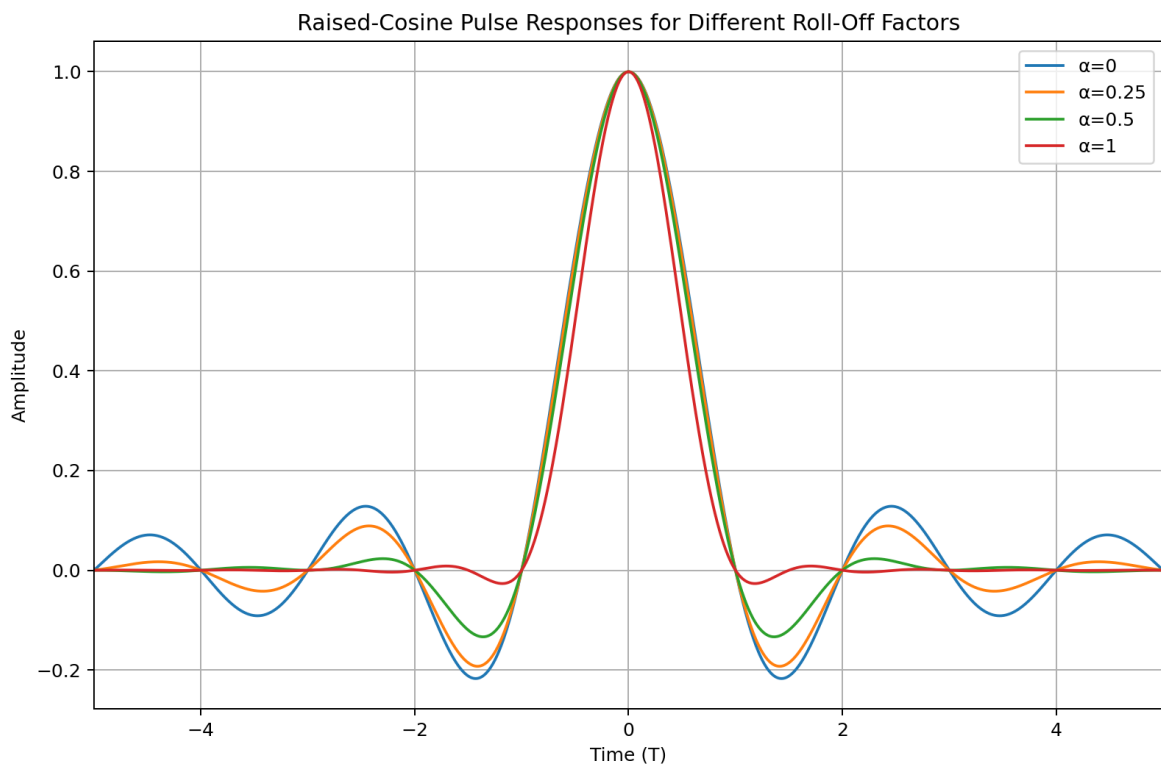
The following graphs were generated from the MATLAB simulation and inserted into the laboratory record. They cover the required pulse responses, frequency responses, pulse-shaped stream, Nyquist validation, RRC cascade, and bandwidth-versus-roll-off plot.

Figure 1: Pulse Responses



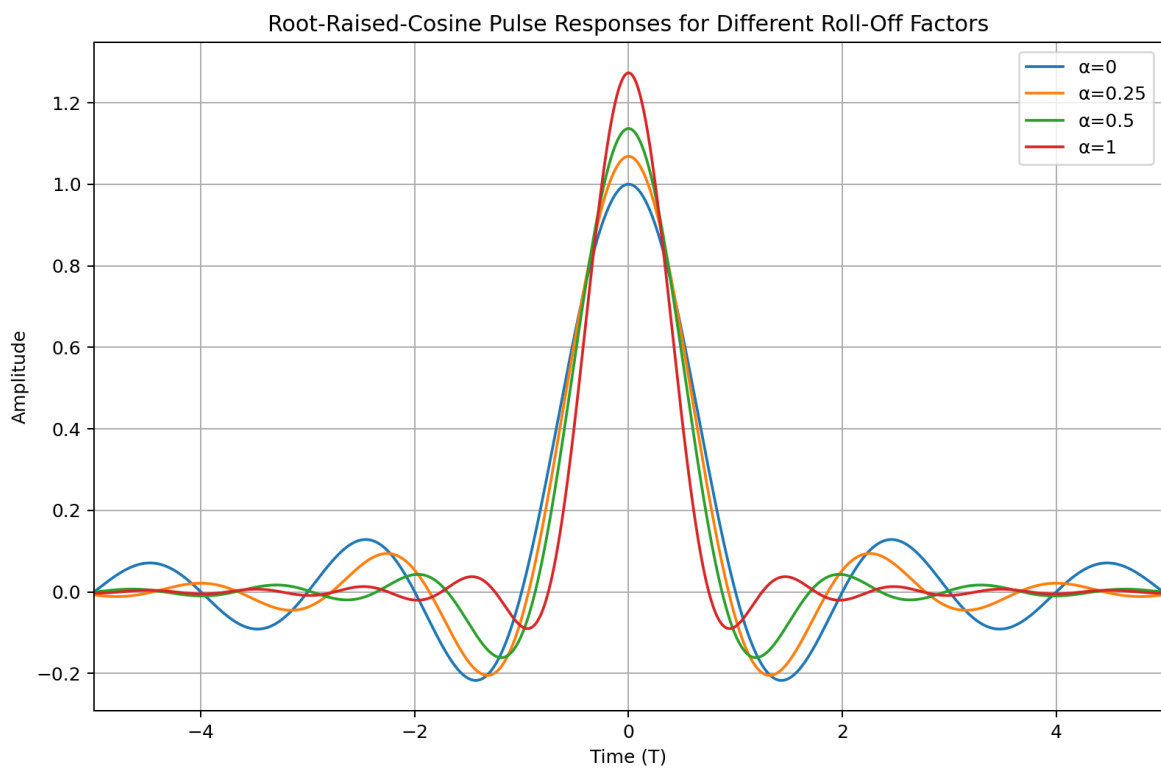
Comparison of rectangular, sinc, raised-cosine and root-raised-cosine pulses.

Figure 2: Raised-Cosine Roll-Off Comparison



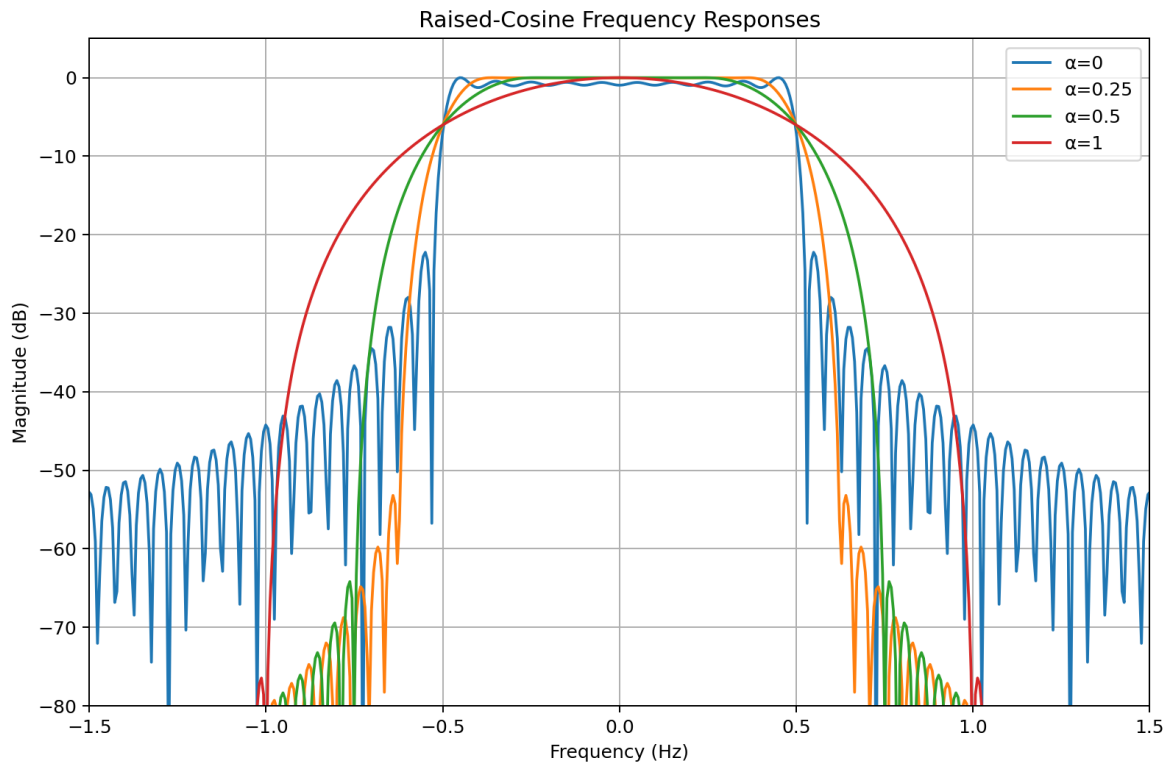
RC responses for $\alpha = 0, 0.25, 0.5$ and 1 .

Figure 3: Root-Raised-Cosine Roll-Off Comparison



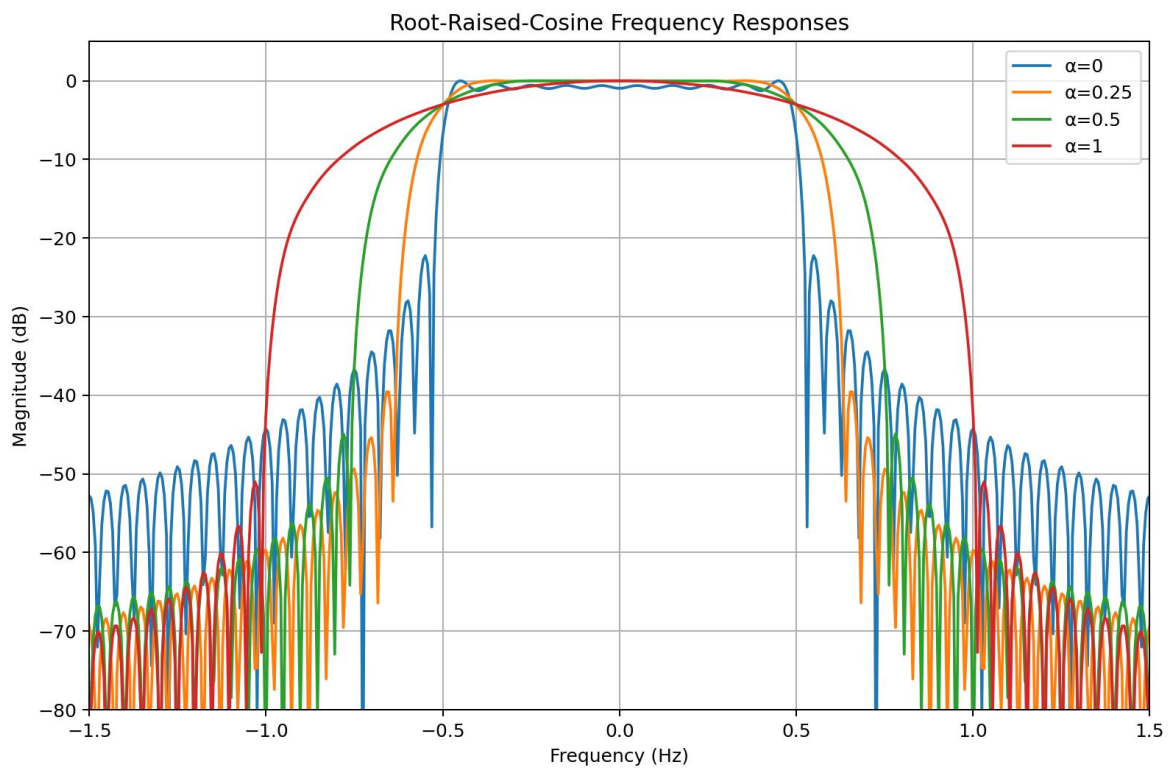
RRC responses for $\alpha = 0, 0.25, 0.5$ and 1 .

Figure 4: Raised-Cosine Frequency Responses



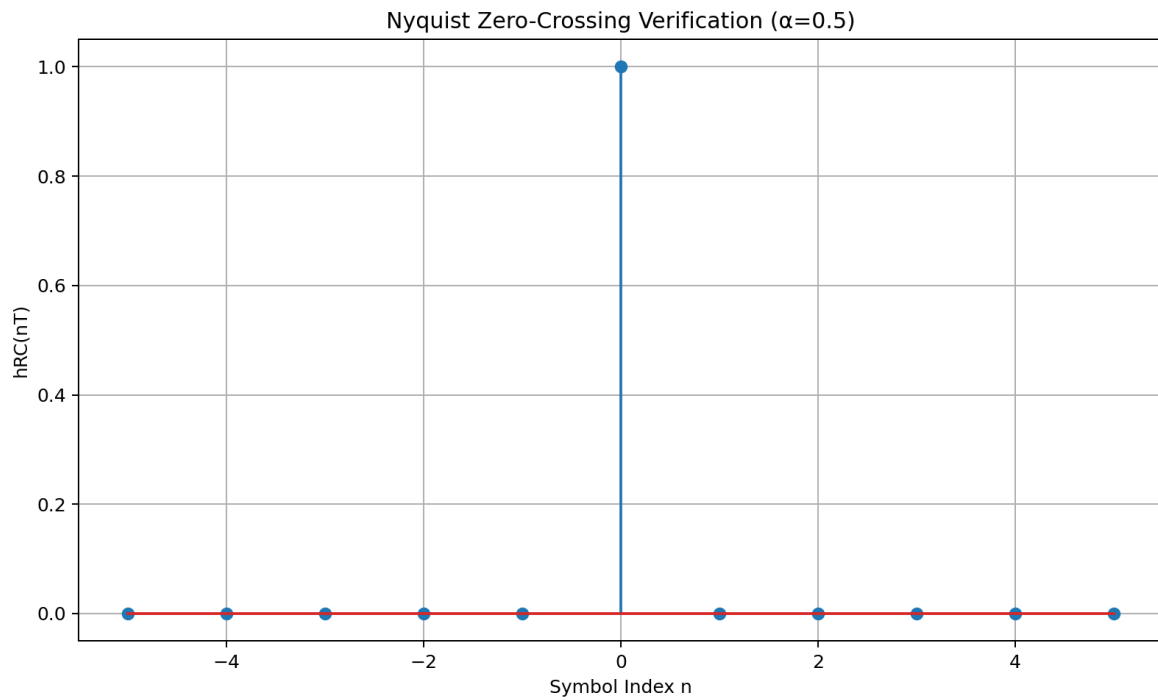
Frequency-domain comparison for the four roll-off factors.

Figure 5: Root-Raised-Cosine Frequency Responses



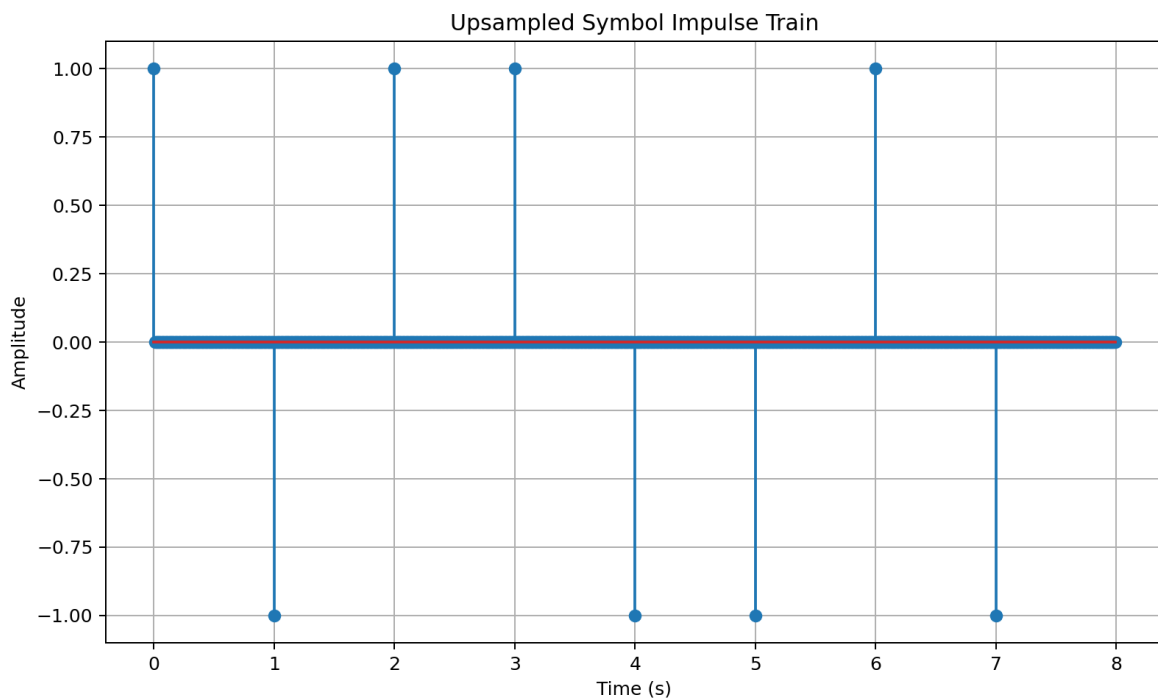
Frequency-domain comparison for the four roll-off factors.

Figure 6: Nyquist Zero-Crossing Verification



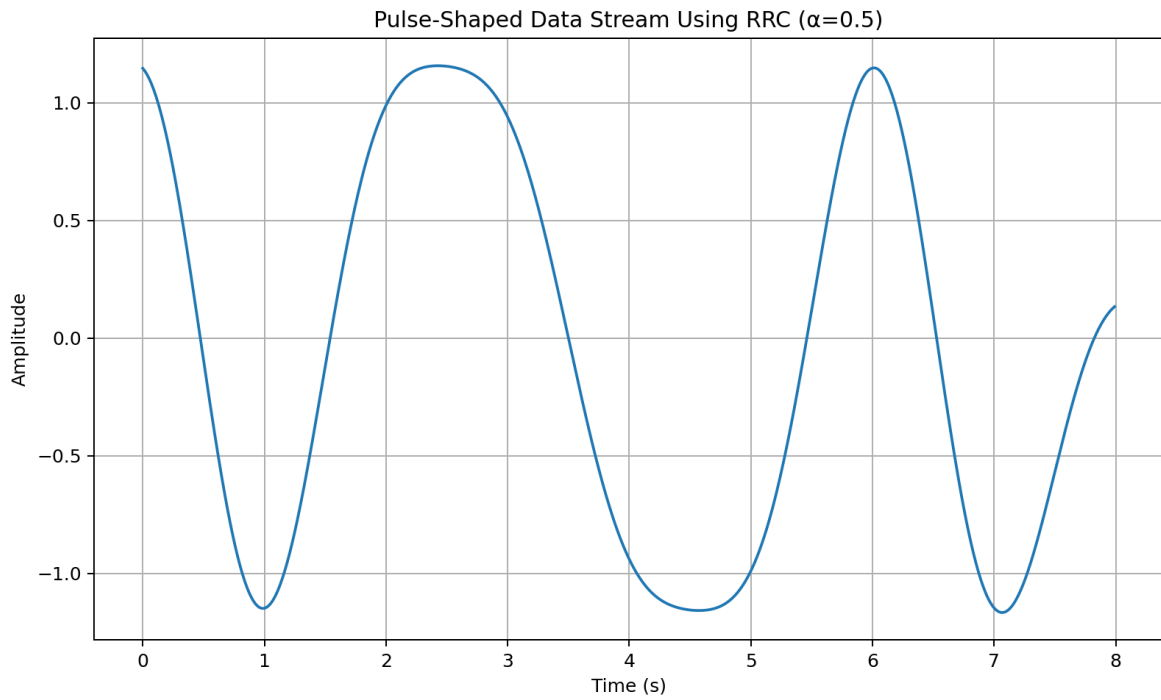
The RC response is approximately zero at all integer symbol intervals except $n = 0$.

Figure 7: Upsampled Symbol Impulse Train



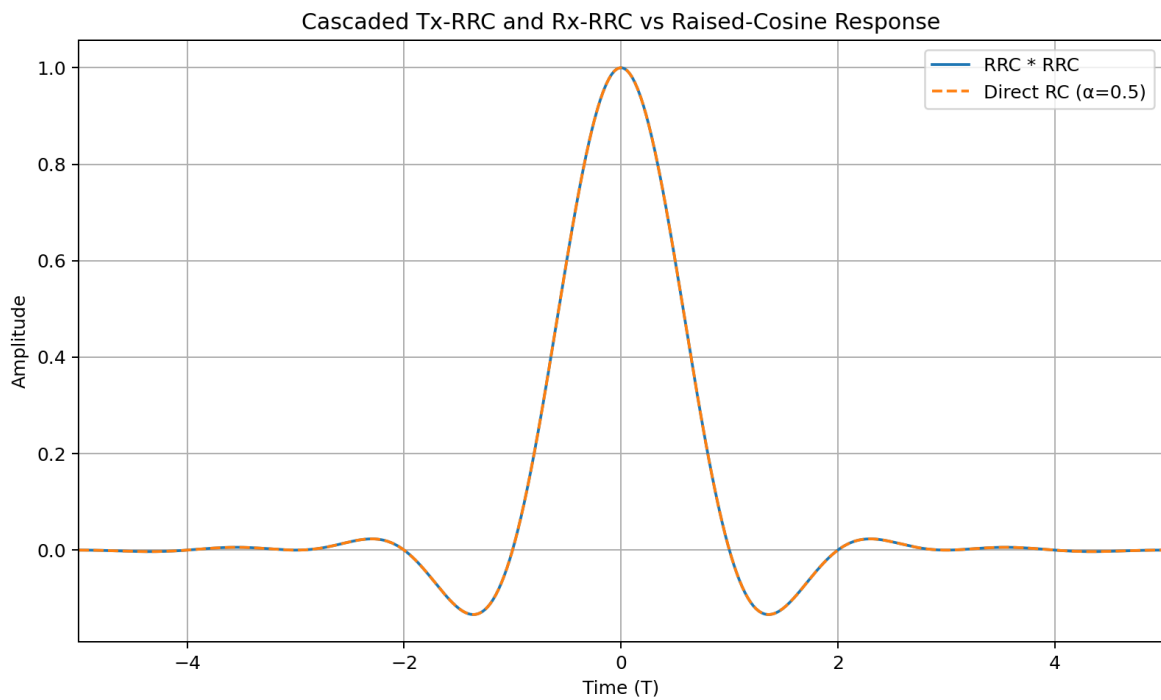
Input symbol sequence after zero-stuffing/upsampling.

Figure 8: Pulse-Shaped Data Stream



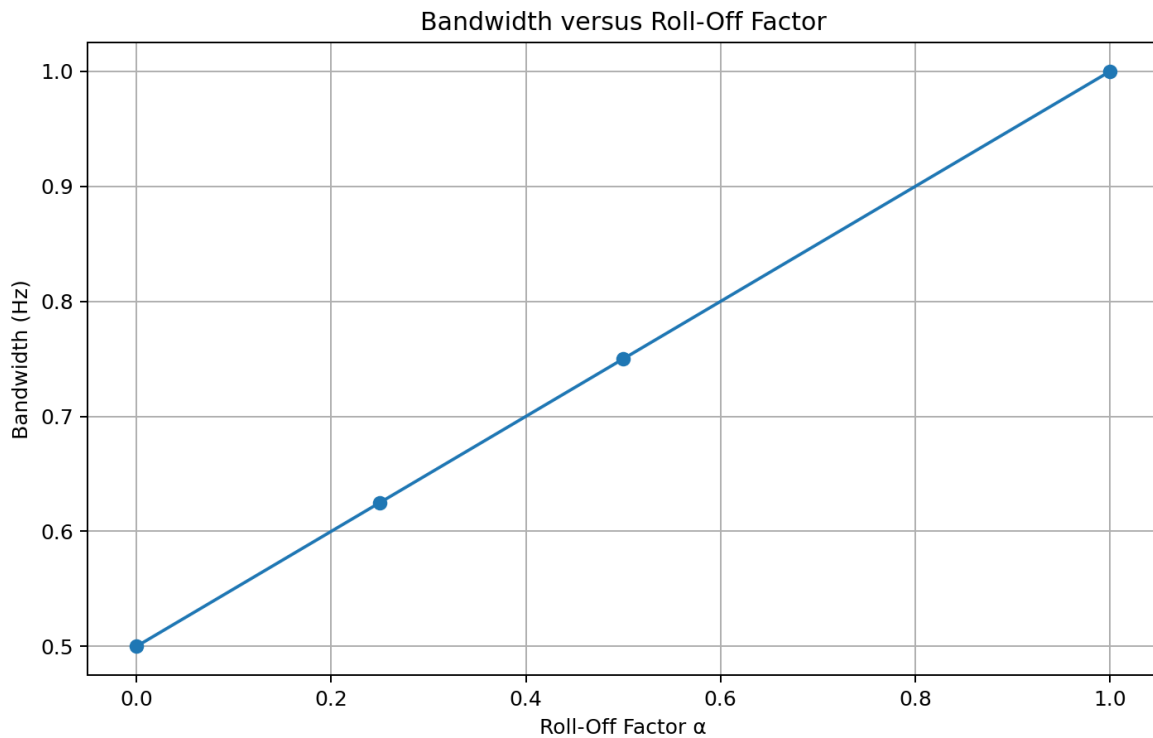
Data stream after RRC pulse shaping.

Figure 9: RRC Cascade vs RC



Cascaded transmitter and receiver RRC response compared with direct RC.

Figure 10: Bandwidth versus Roll-Off Factor



Theoretical bandwidth $B = (1 + \alpha)/(2T)$ for $T = 1$ s.

18. Graph-Based Observation Summary

Graph	Observation	Inference
Pulse responses	Different pulse shapes have different time-domain decay and spectral characteristics.	Pulse shaping controls bandwidth and ISI.
RC roll-off	Increasing α produces a more time-localized pulse.	Higher α improves time localization at the cost of bandwidth.
Frequency response	Higher α gives a wider occupied/transition band.	Bandwidth increases with α .
Zero crossings	$h_{RC}(nT) \approx 0$ for $n \neq 0$ and $h_{RC}(0) \approx 1$.	Nyquist zero-ISI condition is verified.
Pulse-shaped stream	The impulse train becomes a smooth waveform after RRC filtering.	RRC provides practical pulse shaping.
RRC cascade	RRC * RRC closely matches the RC response.	Tx/Rx RRC filters form an overall RC response.
Bandwidth vs α	Bandwidth increases linearly with α .	Confirms $B = (1 + \alpha)/(2T)$.